



## **S.B. JAIN INSTITUTE OF TECHNOLOGY MANAGEMENT & RESEARCH, NAGPUR**

### **Practical 03**

**Aim:** Automate student marksheet generation, system information display, Fibonacci and prime number generation, and file management operations using shell scripts to enhance computational efficiency and user interaction.

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**Date of Performance:**

**Date of Submission:**

❖ **Aim:** Automate student marksheet generation, system information display, Fibonacci and prime number generation, and file management operations using shell scripts to enhance computational efficiency and user interaction.

❖ **Tasks to be done in this Practical.**

- a) Write a shell script to generate mark- sheet of a student. Take 3 subjects, calculate and display total marks, percentage and Class obtained by the student.
- b) Write a menu driven shell script which will print the following menu and execute the given task.
  - Display calendar of current month.
  - Display today's date and time.
  - Display usernames those are currently logged in the system.
  - Display your terminal number
- c) Write a shell script which will generate first n Fibonacci numbers like: 1, 1, 2, 3, 5, 13
- d) Write a shell script which will accept a number b and display first n prime numbers as output.
- e) Write menu driven program for file handling activity
  - Creation of file.
  - Write content in the file.
  - Upend file content.
  - Delete file content

❖ **Objectives:**

1. Automate marksheet generation with total, percentage, and class classification.
2. Develop menu-driven scripts for system information and file operations.
3. Generate Fibonacci and prime numbers for user-defined inputs.

❖ **Requirements:**

✓ **Hardware Requirements:**

- Processor: Minimum 1 GHz
- RAM: 512 MB or higher
- Storage: 100 MB free space

✓ **Software Requirements:**

- Operating System: Linux/Unix-based
- Shell: Bash 4.0 or higher
- Text Editor: Nano, Vim, or any preferred editor



## ❖ Theory:

Shell scripting is a powerful way to automate repetitive tasks and manage system operations efficiently. It allows users to write programs using shell commands and scripting constructs. Shell scripts are interpreted line-by-line by a shell interpreter, making them ideal for administrative tasks, file management, and system automation. This practical encompasses a variety of real-world scenarios that demonstrate the utility of shell scripting for computing tasks and resource management.

### 1. Marksheet Generation

This script takes input marks for three subjects, calculates the total marks, percentage, and determines the class of the student based on predefined conditions. Conditional statements (if-else) are used to classify the performance into distinction, first class, second class, or fail. This exercise emphasizes the use of arithmetic operations and decision-making constructs.

Key concepts include:

- Reading user input using read
- Arithmetic operations with `$((expression))`
- Conditional statements for decision-making

### 2. Menu-Driven Script for System Information

Menu-driven scripts enhance user interaction by presenting a list of options for performing different tasks. In this practical, options are provided to display the calendar of the current month, the current date and time, logged-in users, and the terminal number. The script utilizes looping constructs (while) and case statements for structured flow control.

**Commands used:**

- cal for displaying the calendar
- date for showing current date and time
- who to list logged-in users
- tty to identify the terminal



### 3. Fibonacci Number Generation

Fibonacci numbers are a sequence where each term is the sum of the two preceding ones. The script uses iterative constructs (for loop) to generate n terms based on user input. This practical illustrates the use of loop control and variable swapping to generate series data efficiently.

#### 4. Prime Number Display

This script accepts an integer n and outputs the first n prime numbers. A nested loop checks divisibility to determine if a number is prime. The practical demonstrates logic building for number-theoretic operations using loops and conditionals.

#### 5. Menu-Driven File Management

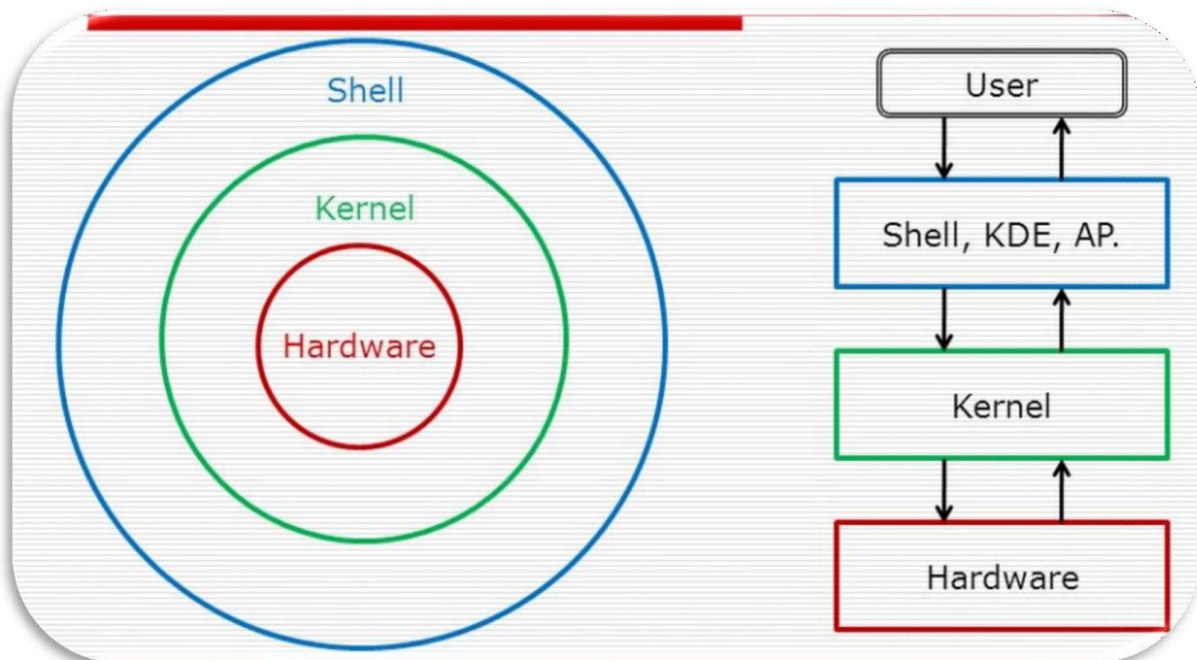
The file handling script enables users to create, write, append, and delete file content. The case construct manages different file operations.

Commands include:

- ❑ touch to create files
- ❑ cat for writing and appending content
- ❑ rm for deleting files

This exercise emphasizes text manipulation, input handling, and file control mechanisms in Unix-like environments.

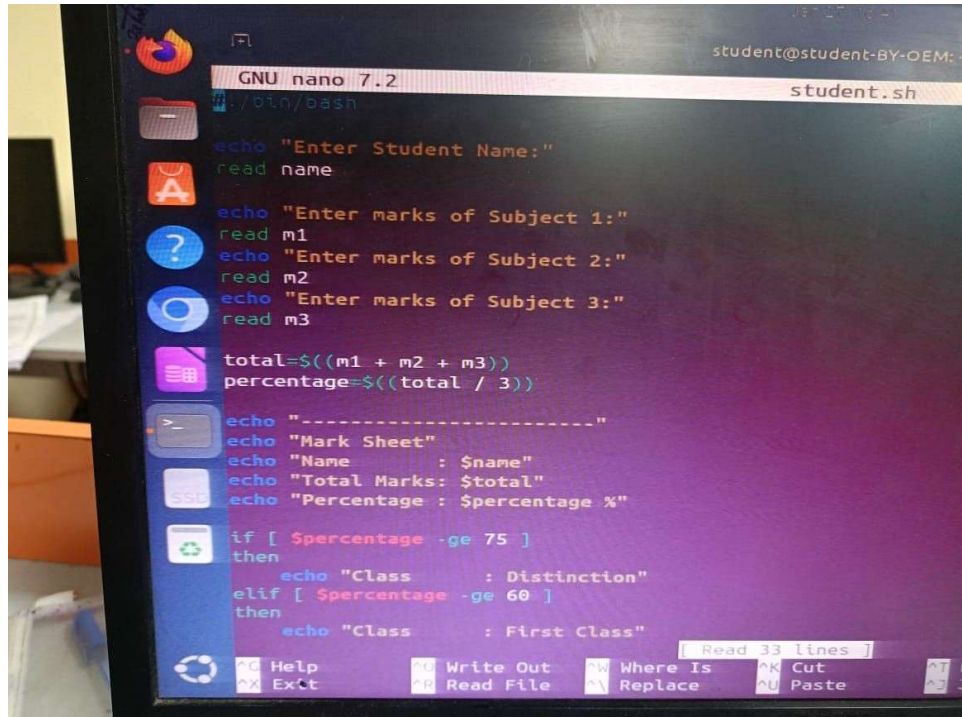
#### Diagrammatical View of Shell



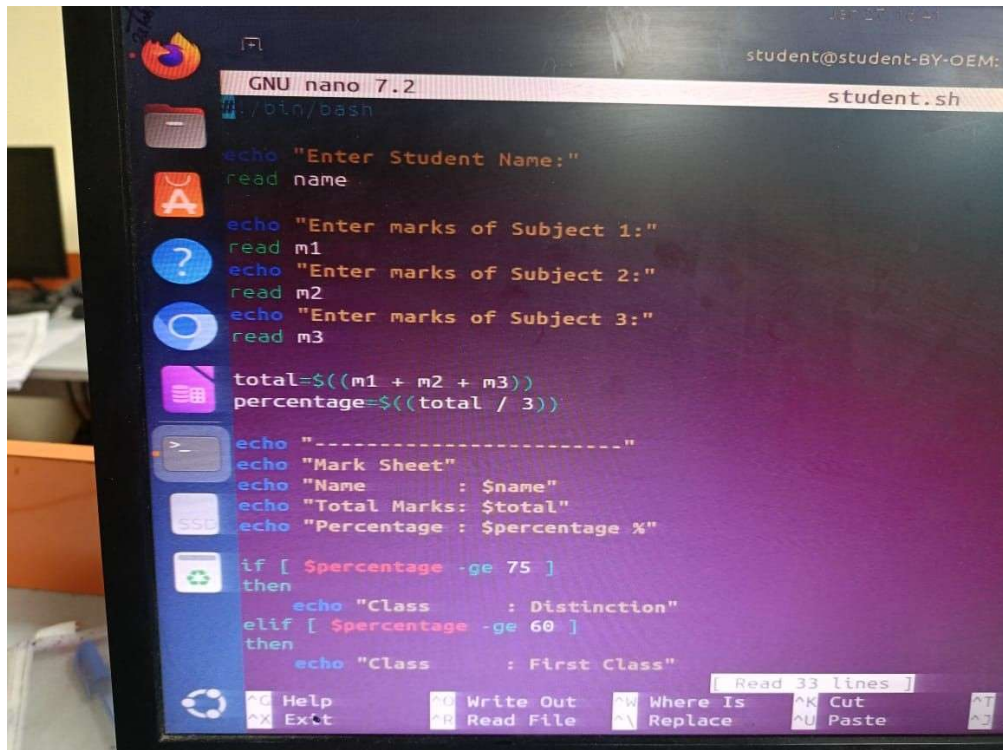
❖ CODES

1. Write a shell script to generate mark- sheet of a student. Take 3 subjects, calculate and display total marks, percentage and Class obtained by the student.

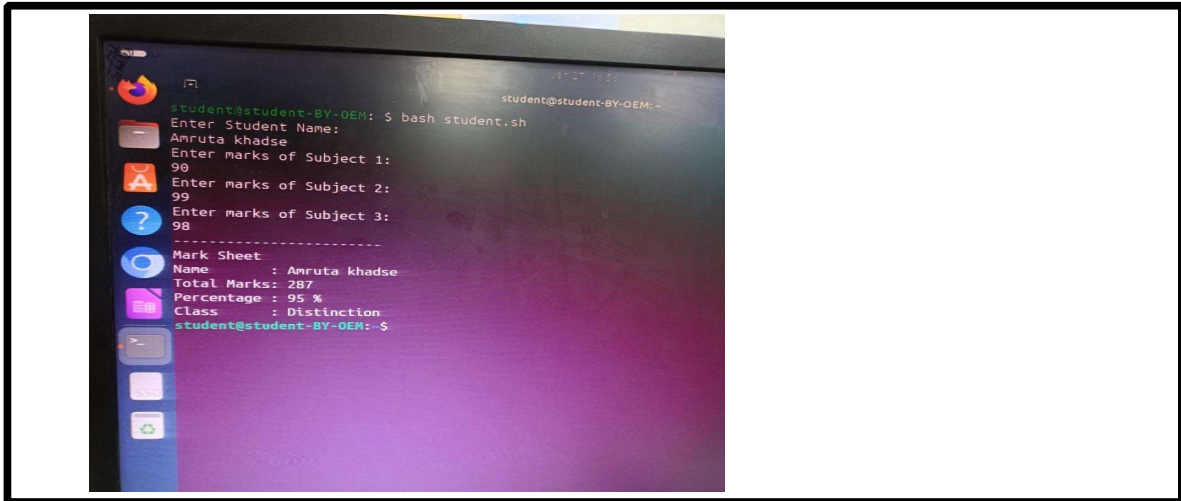
Output 1:



```
GNU nano 7.2 student.sh
./bin/bash
echo "Enter Student Name:"
read name
echo "Enter marks of Subject 1:"
read m1
echo "Enter marks of Subject 2:"
read m2
echo "Enter marks of Subject 3:"
read m3
total=$((m1 + m2 + m3))
percentage=$((total / 3))
echo "-----"
echo "Mark Sheet"
echo "Name      : $name"
echo "Total Marks: $total"
echo "Percentage : $percentage %"
if [ $percentage -ge 75 ]
then
echo "Class      : Distinction"
elif [ $percentage -ge 60 ]
then
echo "Class      : First Class"
```

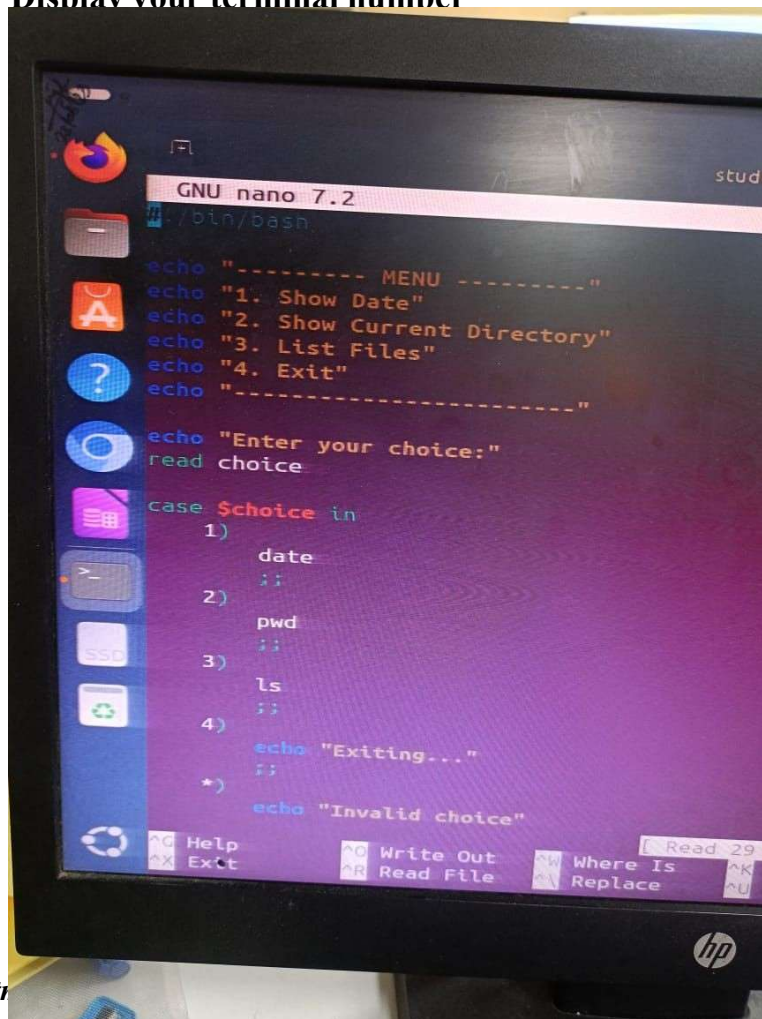


```
GNU nano 7.2 student.sh
./bin/bash
echo "Enter Student Name:"
read name
echo "Enter marks of Subject 1:"
read m1
echo "Enter marks of Subject 2:"
read m2
echo "Enter marks of Subject 3:"
read m3
total=$((m1 + m2 + m3))
percentage=$((total / 3))
echo "-----"
echo "Mark Sheet"
echo "Name      : $name"
echo "Total Marks: $total"
echo "Percentage : $percentage %"
if [ $percentage -ge 75 ]
then
echo "Class      : Distinction"
elif [ $percentage -ge 60 ]
then
echo "Class      : First Class"
```

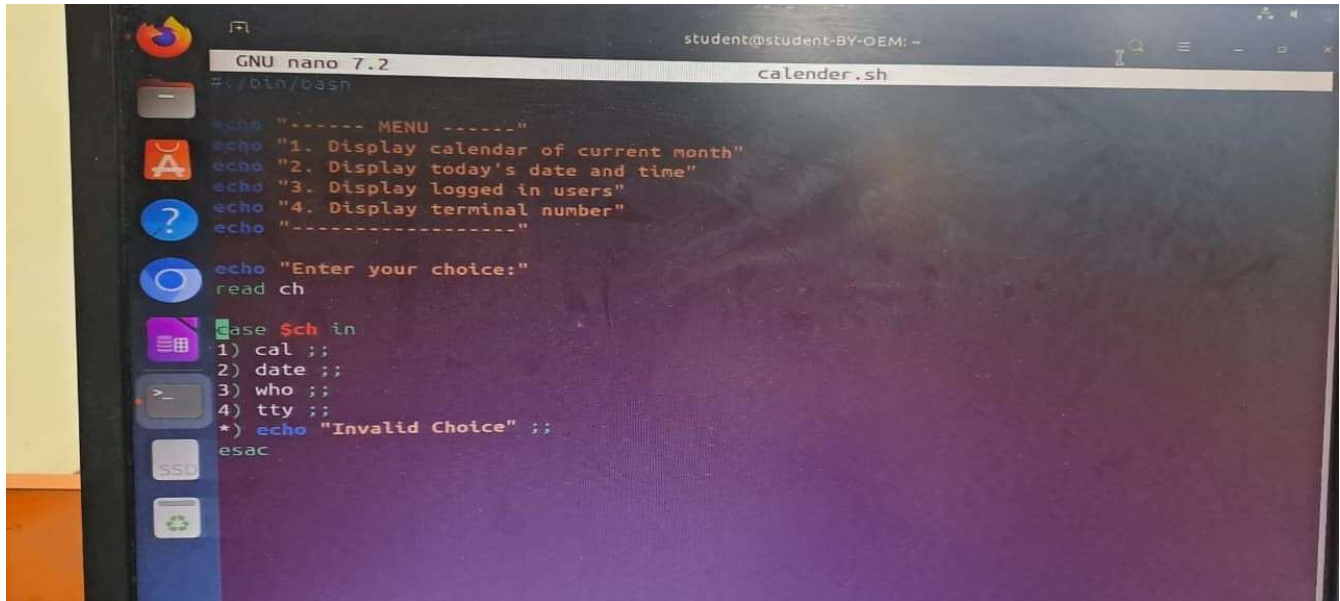


2. Write a menu driven shell script which will print the following menu and execute the given task.

- Display calendar of current month.
- Display today's date and time.
- Display usernames those are currently logged in the system.
- Display your terminal number



**Output 2:**



The screenshot shows a terminal window with the GNU nano 7.2 editor. The file being edited is 'calender.sh'. The script contains a menu with four options: 1. Display calendar of current month, 2. Display today's date and time, 3. Display logged in users, and 4. Display terminal number. The script prompts the user to enter a choice and uses a case statement to execute the corresponding command: 'cal' for option 1, 'date' for option 2, 'who' for option 3, and 'tty' for option 4. Any other choice is considered invalid.

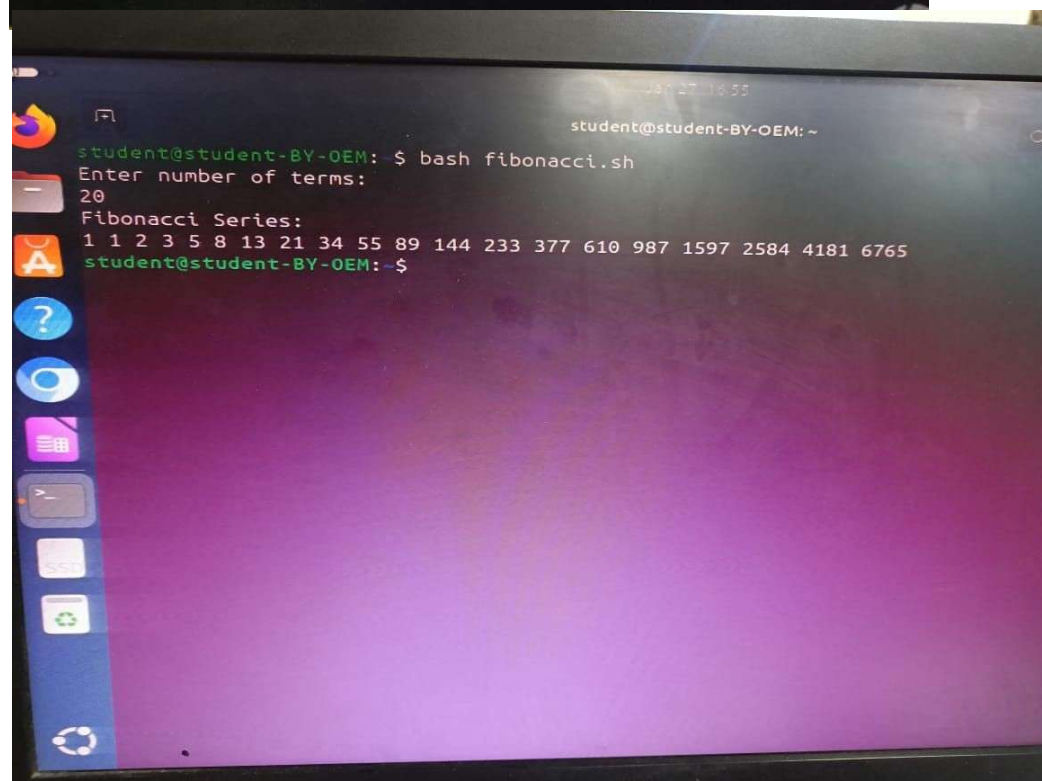
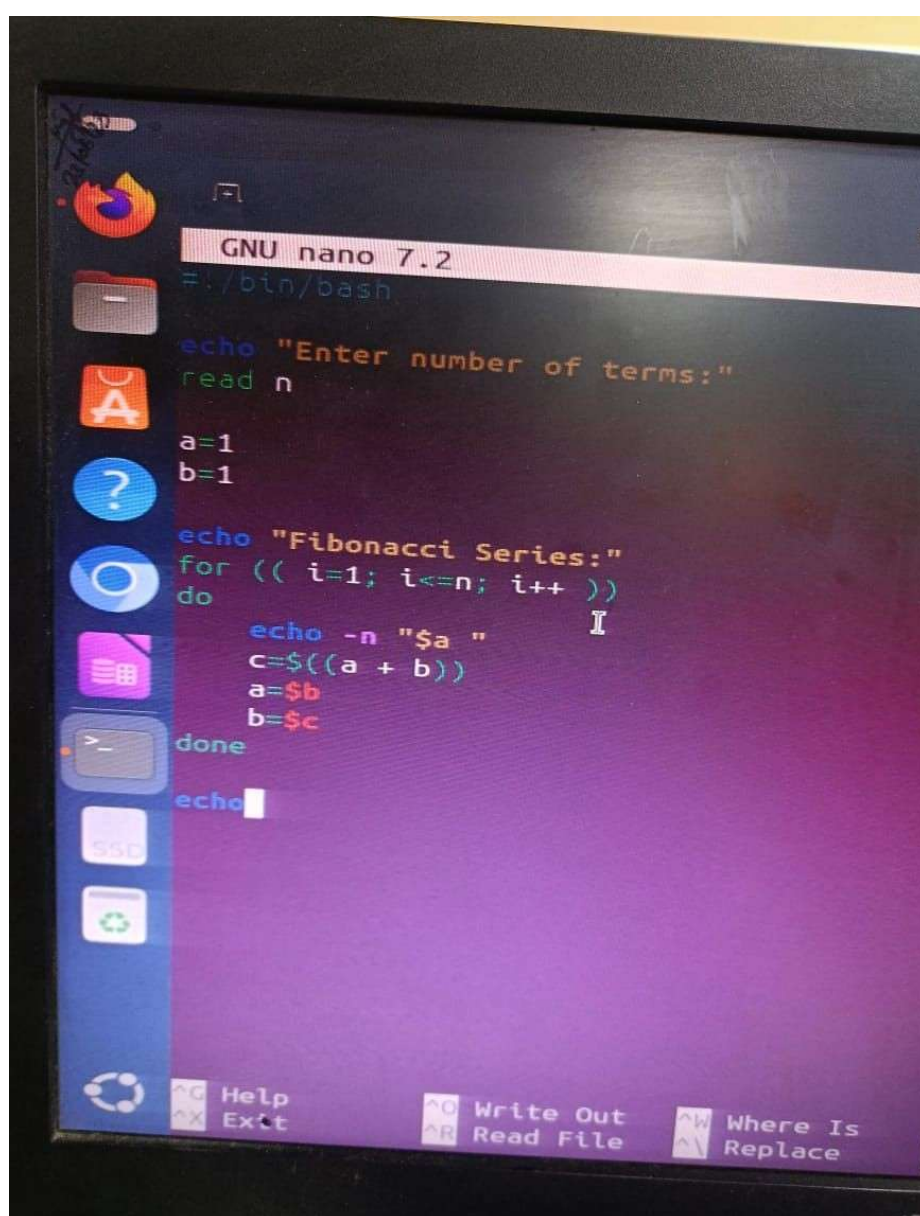
```
GNU nano 7.2 calender.sh
#!/bin/bash

echo "----- MENU -----"
echo "1. Display calendar of current month"
echo "2. Display today's date and time"
echo "3. Display logged in users"
echo "4. Display terminal number"
echo "-----"

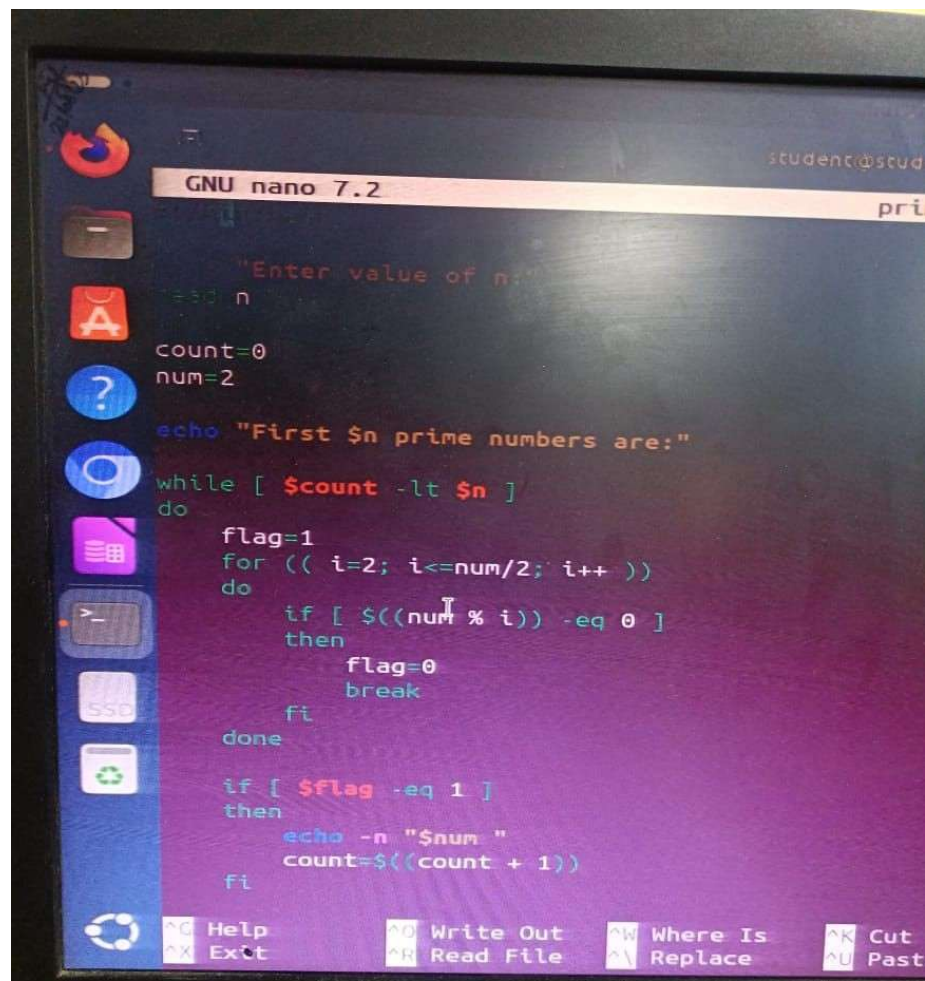
echo "Enter your choice:"
read ch

case $ch in
  1) cal ;;
  2) date ;;
  3) who ;;
  4) tty ;;
  *) echo "Invalid Choice" ;;
esac
```

3. Write a shell script which will generate first n Fibonacci numbers like:  
1, 1, 2, 3, 5, 13

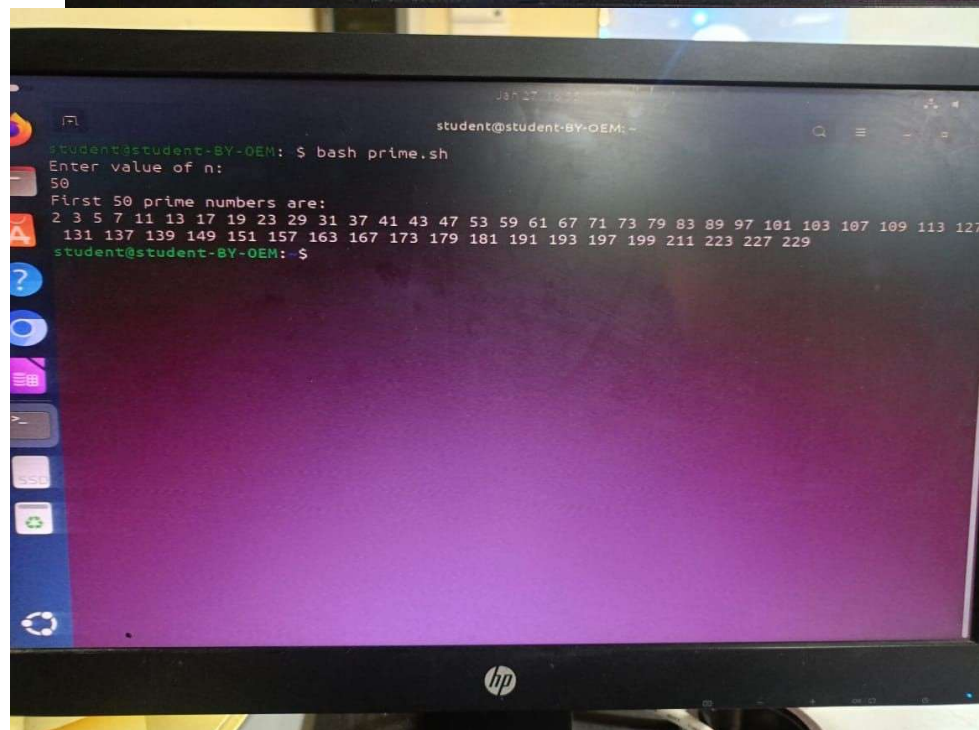


- 4. Write a shell script which will accept a number b and display first n prime numbers as output.**



The screenshot shows a Linux desktop environment with a terminal window open. The terminal is running the GNU nano 7.2 text editor. The script being edited is a shell script that takes an input 'n' and prints the first 'n' prime numbers. The script uses a while loop to iterate through numbers starting from 2, checking for primality using a for loop and a flag variable. The desktop background is dark blue, and the left sidebar contains various application icons including Firefox, Files, and the Dash. The terminal window title bar shows 'student@stud'.

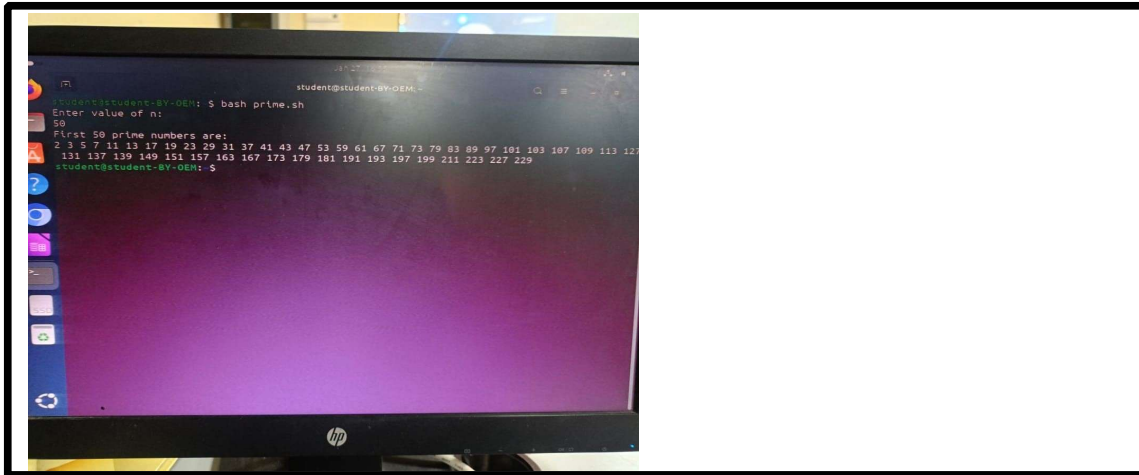
```
GNU nano 7.2 pri
student@stud
"Enter value of n:"
read n
count=0
num=2
echo "First $n prime numbers are:"
while [ $count -lt $n ]
do
    flag=1
    for (( i=2; i<=num/2; i++ ))
    do
        if [ $((num % i)) -eq 0 ]
        then
            flag=0
            break
        fi
    done
    if [ $flag -eq 1 ]
    then
        echo -n "$num "
        count=$((count + 1))
    fi
done
```



The screenshot shows the same Linux desktop environment, but now the terminal window is displaying the output of the script. The user has entered '50' as the value for 'n'. The script has successfully printed the first 50 prime numbers. The terminal window title bar shows 'student@student-BY-OEM: ~'. The desktop background and sidebar are the same as in the previous image.

```
student@student-BY-OEM: ~
student@student-BY-OEM: $ bash prime.sh
Enter value of n:
50
First 50 prime numbers are:
2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89 97 101 103 107 109 113 127
131 137 139 149 151 157 163 167 173 179 181 191 193 197 199 211 223 227 229
student@student-BY-OEM: $
```

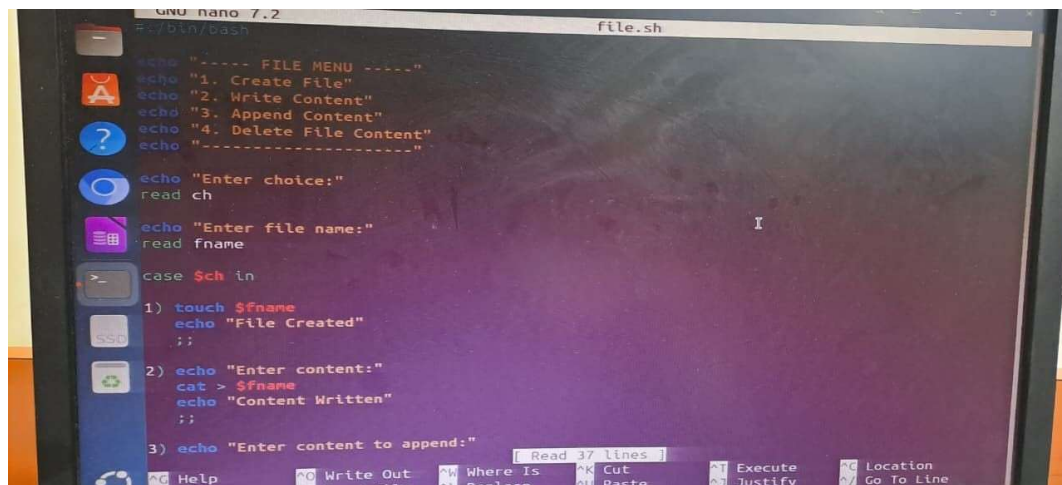
**Output 4:**



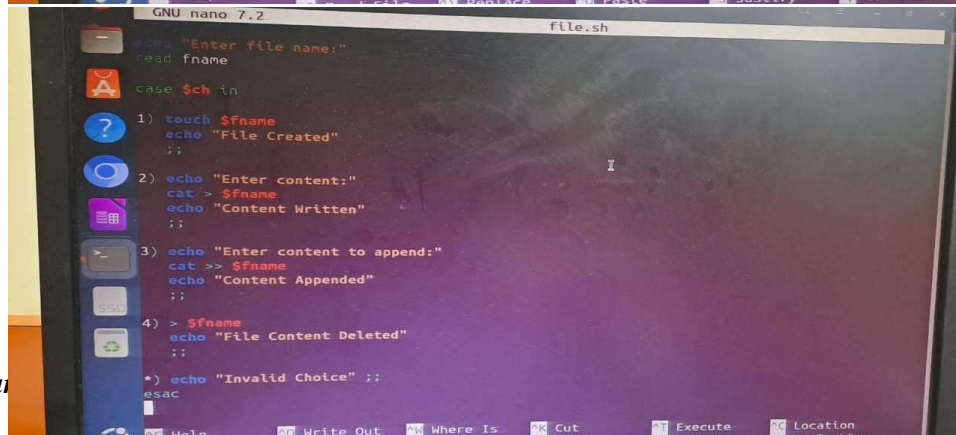
```
student@student-BV-OEM: ~  
student@student-BV-OEM: $ bash prime.sh  
Enter value of n:  
50  
First 50 prime numbers are:  
2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89 97 101 103 107 109 113 127  
131 137 139 149 151 157 163 167 173 179 181 191 193 197 199 211 223 227 229  
student@student-BV-OEM: $
```

5. Write menu driven program for file handling activity

- Creation of file.
- Write content in the file.
- Upend file content.
- Delete file content

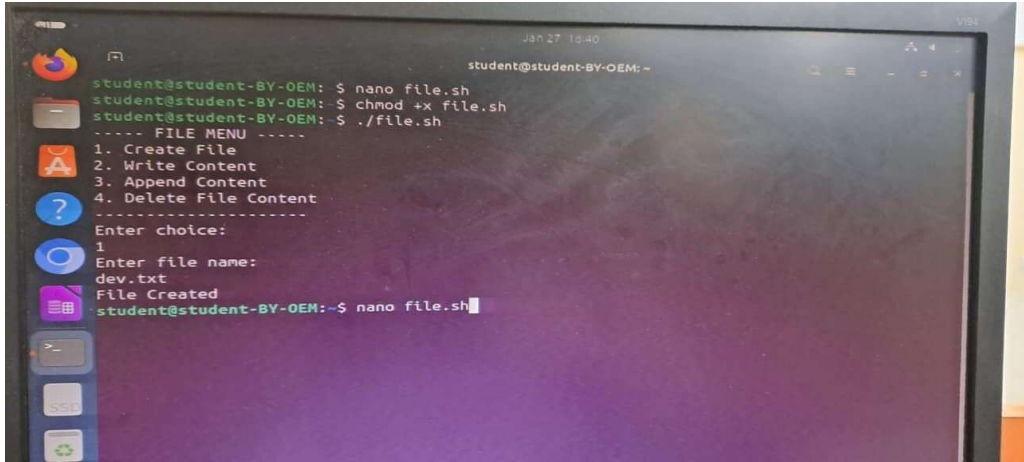


```
GNU nano 7.2 file.sh  
# ./01n/bash  
echo "----- FILE MENU -----"  
echo "1. Create File"  
echo "2. Write Content"  
echo "3. Append Content"  
echo "4. Delete File Content"  
echo "-----"  
echo "Enter choice:"  
read ch  
echo "Enter file name:"  
read fname  
case $ch in  
1) touch $fname  
echo "File Created"  
;;  
2) echo "Enter content:"  
cat > $fname  
echo "Content Written"  
;;  
3) echo "Enter content to append:"  
cat >> $fname  
echo "Content Appended"  
;;  
4) > $fname  
echo "File Content Deleted"  
;;  
*) echo "Invalid Choice" ;;  
esac
```

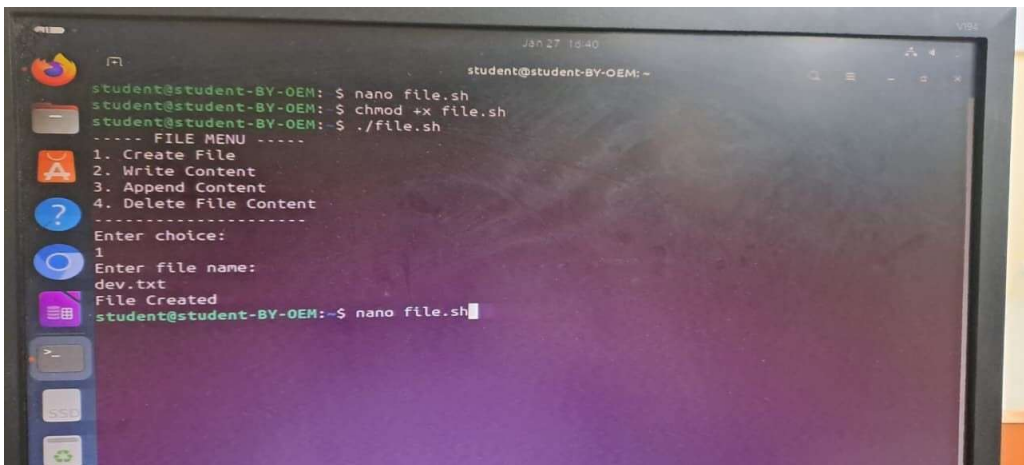


```
GNU nano 7.2 file.sh  
echo "Enter file name:"  
read fname  
case $ch in  
1) touch $fname  
echo "File Created"  
;;  
2) echo "Enter content:"  
cat > $fname  
echo "Content Written"  
;;  
3) echo "Enter content to append:"  
cat >> $fname  
echo "Content Appended"  
;;  
4) > $fname  
echo "File Content Deleted"  
;;  
*) echo "Invalid Choice" ;;  
esac
```

**Output 5:**



```
student@student-BY-OEM: ~
JAN 27 15:40
student@student-BY-OEM: ~
student@student-BY-OEM: $ nano file.sh
student@student-BY-OEM: $ chmod +x file.sh
student@student-BY-OEM: $ ./file.sh
----- FILE MENU -----
1. Create File
2. Write Content
3. Append Content
4. Delete File Content
-----
Enter choice:
1
Enter file name:
dev.txt
File Created
student@student-BY-OEM:~$ nano file.sh
```



```
student@student-BY-OEM: ~
JAN 27 15:40
student@student-BY-OEM: ~
student@student-BY-OEM: $ nano file.sh
student@student-BY-OEM: $ chmod +x file.sh
student@student-BY-OEM: $ ./file.sh
----- FILE MENU -----
1. Create File
2. Write Content
3. Append Content
4. Delete File Content
-----
Enter choice:
1
Enter file name:
dev.txt
File Created
student@student-BY-OEM:~$ nano file.sh
```

❖ **Conclusion:** In this practical, we conclude that shell scripting efficiently automates tasks like marksheet generation, system information display, number computations, and file management, enhancing system operations and user interaction through command-line utilities.

❖ **Discussion Questions:**

1. **What is the purpose of using shell scripting in this practical?**  
**Which command is used to display the current date and time?**
2. **How does the script calculate the Fibonacci sequence?**
3. **Which command is used to create a file in the file management script? Ans:**
4. **How does the prime number script determine if a number is prime? Ans: The**

❖ **References:**

[https://www.tutorialspoint.com/unix/shell\\_scripting.html](https://www.tutorialspoint.com/unix/shell_scripting.html) <https://www.javatpoint.com/shell-scripting-tutorial>

**Date:**     /     /2026

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**Signature**

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Sem: 4 / 2025-26